
CE 551: Geosynthetics and their applications

Course Number: CE 551

Course Name: Geosynthetics and their applications

L-T-P-C: 3-0-0-3

Prerequisites: CE 302 - Geotechnical Engineering

Intended for: UG/PG

Distribution: Discipline Core Semester: Odd/Even

Approval: 9th Senate

Course Contents

- **An Overview of Geosynthetics:** Description of Geosynthetics, Properties, Functions. [4 Lectures]
- **Properties and tests:** Properties of geosynthetics, standard testing procedures based on applications. [8 Lectures]
- **Soil Reinforcement:** Mechanism, Reinforced slopes, Embankments on soft ground, Reinforced Embankments, Reinforced soil walls and Slope stabilization. [9 Lectures]
- **Geosynthetics for Highways:** Roadway Reinforcement, Separation, Filtration, Drainage, Reinforcement, Moisture Barrier, Membrane encapsulation. [6 Lectures]
- **Ground Improvement applications:** Drainage, PVDs, French Drains, encapsulated sand tubes, on-shore applications. [7 Lectures]
- **Geoenvironmental Applications:** Geomembranes for landfills and ponds, Geosynthetic clay liners, designing with GCL's, Filtration, Erosion control, Slope protection. [8 Lectures]

Text books:

1. R. M. Koerner, **Designing with Geosynthetics**, 6th edition, Xlibris Corporation, 2012.
2. G L Sivakumar Babu, **An Introduction to Soil Reinforcement and Geosynthetics**, Universities Press, 2006.

Reference Books:

1. G. V. Rao and G. V. S. S. Raju, **Engineering with Geosynthetics**, McGraw Hill Education India Pvt Ltd., 1998.
2. Sanjay Kumar Shukla, **Geosynthetics and their applications**, Thomas Telford Publications, 2002.